

|         |                  |               |                                                |
|---------|------------------|---------------|------------------------------------------------|
| Date    | 15 February 2025 | Maximum Marks | 4 Marks                                        |
| Team ID | SWTID-2026-5019  | Project Name  | Import Data using Transform Maps (Spreadsheet) |

PROJECT OVERVIEW & CONTEXT

This micro project focuses on importing structured data from an external spreadsheet into the ServiceNow platform using **Import Sets** and **Transform Maps**. The project simulates a real-world scenario where bulk employee data is received in Excel format and must be migrated into ServiceNow efficiently and accurately. The spreadsheet data is first loaded into an Import Set table, which serves as a staging area, and a Transform Map is then configured to map source fields to the appropriate target table fields (sys\_user). Upon executing the transformation, records are created in the target table and validated to ensure data integrity. This project helps beginners understand the end-to-end data import process in ServiceNow, including data loading, field mapping, transformation execution, and result verification.

**Solution Architecture:** Solution architecture is a structured engineering process that bridges the gap between business operational problems and technology implementations. It establishes technical blueprints to govern data flows, boundaries, transform behaviors, and auditability.

- 1 **Solve Existing Business Problems**

Deliver an automated, error-free ETL data migration pipeline that eliminates manual data entry, prevents schema corruption, and accelerates onboarding.
- 2 **Describe Structure & Behavior**

Provide stakeholders with clear architectural models illustrating data quarantine, field mapping transformations, coalesce decisions, and target storage.
- 3 **Define Features & Requirements**

Specify staging schemas (u\_imp\_\*), coalesce key matching, reference field resolvers, and row-level validation constraints.
- 4 **Provide Delivery Specifications**

Establish exact technical execution runbooks under which the migration solution is configured, tested in sub-production, executed, and audited.

| Architectural Layer         | ServiceNow Component            | Functional Responsibility & Design Decisions                                                                                        |
|-----------------------------|---------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| 1. Ingestion Layer          | Data Source (File Loader)       | Ingests external Excel spreadsheet (.xlsx), parses row headers, and handles sheet delimiter definitions.                            |
| 2. Staging Layer            | Import Set Table (u_imp_*)      | Quarantines uncleaned raw rows in an isolated buffer, preventing direct contact with live production database tables.               |
| 3. Transformation Layer     | Transform Map & Coalesce Engine | Executes 1:1 field mappings, runs onBefore sanitization scripts, and determines Insert vs. Update actions via unique coalesce keys. |
| 4. Target Persistence Layer | Target Table (sys_user)         | Commits validated employee records into production, binding references to Department (cmn_department) and Location.                 |
| 5. Observability Layer      | Transform History & Import Logs | Records row-by-row telemetry (inserted, updated, skipped, error) and generates execution audit trails for data integrity sign-off.  |

# ServiceNow Solution Architecture Diagram

End-to-End ETL Pipeline Architecture: External Spreadsheet to Target sys\_user Table

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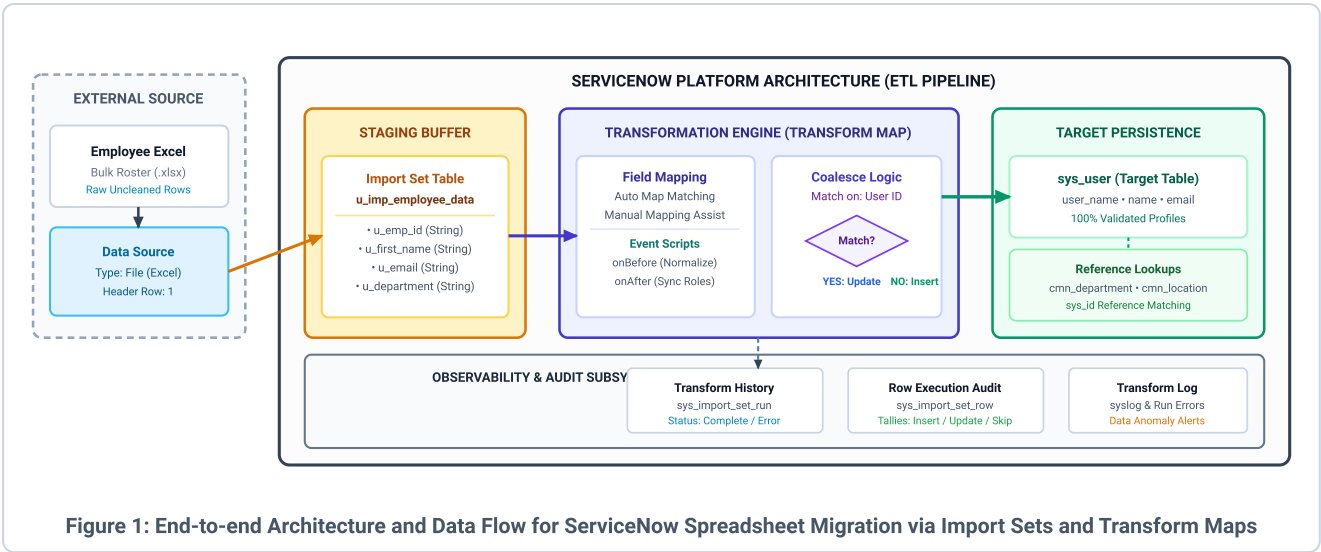


Figure 1: End-to-end Architecture and Data Flow for ServiceNow Spreadsheet Migration via Import Sets and Transform Maps

| Step | Architecture Phase     | Engine Module              | Detailed Operational Execution & Verification Rule                                                                                                                         |
|------|------------------------|----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1    | Data Ingestion         | Load Data / Data Source    | Uploads the Excel file (.xlsx), sets header row indexing, and parses rows into the staging table (u_imp_employee_data). Quarantines data without touching production.      |
| 2    | Schema Mapping         | Transform Map & Field Maps | Executes 'Auto Map Matching Fields' for common attributes (First Name, Last Name, Email) and manual mapping assist for custom keys (u_emp_id → user_name).                 |
| 3    | Identity Deduplication | Coalesce Engine            | Sets Coalesce = true on User ID / Email. Evaluates incoming rows against target records: updates existing profiles and inserts new profiles with zero duplicate accounts.  |
| 4    | Data Transformation    | Event Scripts (onBefore)   | Executes script hooks to format non-standard date strings into YYYY-MM-DD and dynamically queries sys_id references for Department and Location fields.                    |
| 5    | Audit & Verification   | Transform History Log      | Audits the completed Transform History (sys_import_set_run) verifying total inserts, updates, and zero failed rows. Cross-checks live sys_user table to confirm integrity. |